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SPECIFICATION

# Calypso Terminal API

## Generic Card

*Version 2.0.0-SNAPSHOT, 2026-07-20*

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***Warning. The present document cancels and replaces the previous release.***

# Link and Contact



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# 1. Abstract

This document is the official technical specification of the **Terminal Generic Card API** standardised by the **Calypso Networks Association** (CNA). It defines, in a language-agnostic way, the interfaces, classes, enumerations, exceptions, structural relationships and behavioural requirements that any conforming implementation of the Terminal Generic Card API MUST satisfy.

The Terminal Generic Card API is part of the broader CNA **Terminal API** family. It exposes the minimal contract required to exchange raw APDUs with an [ISO/IEC 7816-4](#) card that has been selected through the **Terminal Reader API**, without prescribing any specific card application model.

## Document Status

|                   |                                                                                                                                                           |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reference         | YYMMDD-SP-CNATerminalAPI-GenericCard                                                                                                                      |
| Short name        | CNA-TGC-API                                                                                                                                               |
| Version           | 2.0.0-SNAPSHOT                                                                                                                                            |
| Revision date     | 2026-07-20                                                                                                                                                |
| Editor            | Calypso Networks Association                                                                                                                              |
| Source repository | <a href="https://github.com/calypsonet/calypsonet-terminal-genericcard-uml-api">https://github.com/calypsonet/calypsonet-terminal-genericcard-uml-api</a> |
| Reference license | Creative Commons Attribution-NoDerivatives 4.0 International (CC BY-ND 4.0)                                                                               |



The present document specifies the 2.0.0-SNAPSHOT of the Terminal Generic Card API. It defines a single, coherent baseline against which conforming implementations are evaluated; the **Since** column indicates the version in which each member was introduced.

## Revision List



This section lists the high-level changes per version. The complete, fine-grained changelog is maintained in the [CHANGELOG.md](#) file at the root of the source repository.

| Version / Date                       | Modifications |
|--------------------------------------|---------------|
| <b>v2.0.0-SNAPSHOT</b><br>2026-07-20 | Baseline.     |

## 2. Introduction

### 2.1. Purpose

The Terminal Generic Card API defines the contract used by terminal applications that need to exchange raw APDUs with an ISO/IEC 7816-4-compliant card without relying on a card-specific application layer (e.g. Calypso). It complements the **Terminal Reader API** ([CNA-TR-API](#)) by providing:

- a card-selection extension that allows additional successful status words to be declared for the **Select Application** APDU,
- a minimal transaction manager that prepares APDU commands, attaches an optional command identifier and an optional per-command duration bound (relay-attack countermeasure), and processes them as a single batch, returning the raw responses to the application.

Conforming implementations **MUST** guarantee interoperability with reader implementations of the [CNA-TR-API](#) specification.

### 2.2. Scope

This specification covers:

- the factory used to instantiate the public types of the API,
- the card-selection extension used to enrich the selection scenario for generic cards,
- the transaction manager used to prepare and exchange APDUs with the selected card, including the per-command identifier and the per-command duration bound,
- the property holder exposing the API version.

The following topics are **out of scope**:

- the parsing of APDU payloads—the API exposes raw bytes only,
- the application-level dialog with specific card products (covered by dedicated specifications),
- the management of secure channels.

### 2.3. Conformance

A software product conforms to this specification if and only if:

1. it implements every interface and class defined in [Chapter 5](#) with the operations and semantics prescribed in this document,
2. it preserves the structural relationships described in [Chapter 4](#).

Throughout this specification, the key words **MUST**, **MUST NOT**, **SHOULD**, **SHOULD NOT**, **RECOMMENDED**, **MAY** and **OPTIONAL** are to be interpreted as described in [RFC 2119](#) and [RFC 8174](#) when, and only when, they appear in all capitals.

In conformance with [RFC 2119](#), **SHALL** is equivalent to **MUST**; this specification uses **MUST** exclusively in order to avoid ambiguity.

## 2.4. Document Conventions

### 2.4.1. Naming

Type names follow **UpperCamelCase**; operation, parameter and enumeration-member names follow **lowerCamelCase** except for enumeration values which use **UPPER\_SNAKE\_CASE**. Names defined by this specification are reproduced as-is in the UML class diagram ([Section 4.3](#)).

### 2.4.2. Language-agnostic types

The following symbolic type names are used in operation signatures and are mapped, in each binding, to the closest equivalent native type. Implementations **MUST** preserve the semantic intent rather than the syntactic form.

| Symbolic type        | Description                                                    | Typical bindings                             |
|----------------------|----------------------------------------------------------------|----------------------------------------------|
| <b>string</b>        | Sequence of characters, UTF-8 encoded by default.              | <b>String</b> , <b>string</b> , <b>str</b> . |
| <b>boolean</b>       | Two-state truth value.                                         | <b>boolean</b> , <b>bool</b> .               |
| <b>byte</b>          | Signed 8-bit value.                                            | <b>byte</b> , <b>i8</b> .                    |
| <b>integer</b>       | Signed 32-bit integer.                                         | <b>int</b> , <b>Int32</b> .                  |
| <b>long</b>          | Signed 64-bit integer.                                         | <b>long</b> , <b>Int64</b> .                 |
| <b>byte array</b>    | Ordered sequence of octets.                                    | <b>byte[]</b> , <b>[u8]</b> , <b>bytes</b> . |
| <b>list&lt;T&gt;</b> | Ordered collection of <b>T</b> values, may contain duplicates. | <b>List&lt;T&gt;</b> , <b>Vec&lt;T&gt;</b> . |
| <b>void</b>          | Indicates that an operation returns no value.                  | <b>void</b> , <b>Unit</b> , <b>None</b> .    |

### 2.4.3. Type tables

Each interface, class, enumeration and exception defined in this document is described by a table of the form:

|                   |                                                                                                                             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>       | The type's normalized name.                                                                                                 |
| <b>Kind</b>       | The category of the type (interface, class, enumeration, exception, etc.).                                                  |
| <b>Namespace</b>  | The namespace in which the type is defined.                                                                                 |
| <b>Extends</b>    | The type extended by this type, when applicable.                                                                            |
| <b>Implements</b> | The interface implemented by this type, when applicable.                                                                    |
| <b>Since</b>      | The version of the API in which the type was introduced.                                                                    |
| <b>Purpose</b>    | A normative description of the type's responsibility.                                                                       |
| <b>See also</b>   | Cross-references to related operations or types, each a hyperlink to the corresponding table. Present only when applicable. |

The **Purpose** row states, in one or two sentences, the responsibility of the type and the way an instance is obtained. It is meant to be scanned, not read: any content that requires structure or depth—rules, lists, notes, value tables or examples—is placed as prose below the table.

### 2.4.4. Operation tables

Each operation defined in this document is described by a table of the form:

|                  |                                              |
|------------------|----------------------------------------------|
| <b>Signature</b> | The operation's language-agnostic signature. |
|------------------|----------------------------------------------|

|             |                                                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------------------------------|
| Since       | The version of the API in which the operation was introduced.                                                               |
| Description | A normative description of the operation's behaviour.                                                                       |
| Parameters  | A description of each input parameter.                                                                                      |
| Returns     | A description of the returned value, if any.                                                                                |
| Throws      | A list of exceptional conditions, each mapped to a specific exception type.                                                 |
| See also    | Cross-references to related operations or types, each a hyperlink to the corresponding table. Present only when applicable. |

### 2.4.5. Operation contracts

To avoid repetition, the following rules apply to every operation table in [Chapter 5](#) and are therefore not restated for each operation:

- **Non-null arguments.** Unless explicitly stated otherwise, every input parameter **MUST** be non-**null**. An implementation **MUST** raise **IllegalArgumentException** if a **null** value is supplied. This implicit **null** check is not repeated in the **Throws** row, which states **None**, when no other exception can occur.
- **Non-null results.** Unless the return type is marked nullable (**T?**) or the **Returns** row states otherwise, an operation returns a non-**null** value.
- **Collections.** Unless the return type is marked nullable (**T?**), an operation whose return type is a collection (e.g. **List**, **set**, **map**) returns an empty collection rather than **null**.
- **Fluent composition.** Builder-style operations return the current instance so that calls can be chained.

### 2.4.6. Concurrency

Unless explicitly stated otherwise, the stateful types defined by this specification are **not** thread-safe. A given instance **MUST** be accessed from a single thread at a time; concurrent use of the same instance without external synchronisation results in undefined behaviour. Distinct instances **MAY** be used concurrently.

## 2.5. References and Resources

### 2.5.1. Calypso References

References to CNA Terminal APIs designate the indicated **major** version; any later release within the same major is backward-compatible per that API's evolution policy.

| Reference    | Document                                                                                                       |
|--------------|----------------------------------------------------------------------------------------------------------------|
| [CNA-TR-API] | <a href="#">CNA Terminal API—Reader</a> (SP-CNATerminalAPI-Reader), version 3.0, Calypso Networks Association. |

### 2.5.2. Normative References

| Reference      | Document                                                                                           |
|----------------|----------------------------------------------------------------------------------------------------|
| [RFC 2119]     | <a href="#">Key words for use in RFCs to Indicate Requirement Levels</a> , S. Bradner, March 1997. |
| [RFC 8174]     | <a href="#">Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words</a> , B. Leiba, May 2017.    |
| [ISO/IEC 7816] | <a href="#">Identification cards—Integrated circuit cards.</a>                                     |

### 2.5.3. External Resources

| Resource                     | Link                                                                                            |
|------------------------------|-------------------------------------------------------------------------------------------------|
| Calypso Networks Association | <a href="https://calypsonet.org">https://calypsonet.org</a>                                     |
| Terminal APIs website        | <a href="https://terminal-api.calypsonet.org">https://terminal-api.calypsonet.org</a>           |
| Terminal APIs documentation  | <a href="https://docs.terminal-api.calypsonet.org">https://docs.terminal-api.calypsonet.org</a> |



## 3. Glossary and Acronyms

### 3.1. Glossary

| Term               | Definition                                                                                                                                                 |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| APDU               | Application Protocol Data Unit, the message format used between the terminal and the card, as defined in ISO/IEC 7816-3.                                   |
| Card Reader        | Hardware or software component capable of detecting cards and exchanging data with them.                                                                   |
| Command Identifier | Application-supplied integer attached to a prepared APDU, used to retrieve the corresponding response or to identify the failing command in case of error. |
| Relay Attack       | Software-level attack consisting in interposing a relay between the card and the terminal to redirect APDU exchanges to a distant card.                    |

### 3.2. Acronyms

| Abbreviation | Expansion                                      |
|--------------|------------------------------------------------|
| AID          | Application IDentifier                         |
| APDU         | Application Protocol Data Unit                 |
| CNA          | Calypso Networks Association                   |
| ISO          | International Organization for Standardization |
| SW           | Status Word                                    |

## 4. Architectural Overview

### 4.1. Functional positioning

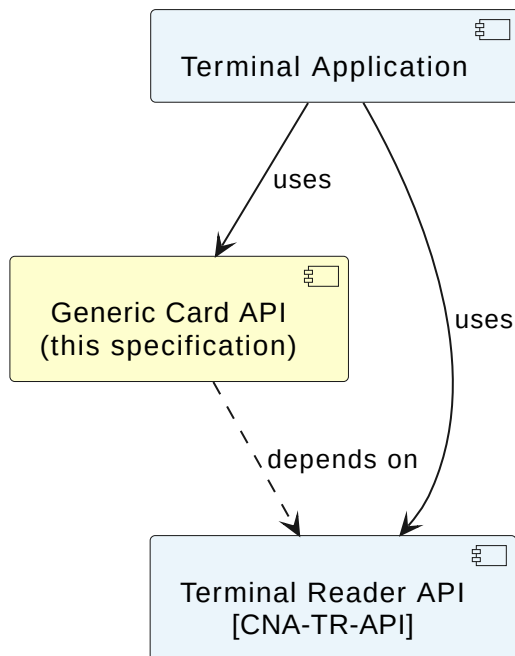


Figure 1. Terminal Generic Card API—Architecture Overview

### 4.2. Logical namespaces

| Namespace   | Role       | Summary                                                           |
|-------------|------------|-------------------------------------------------------------------|
| genericcard | Public API | Factory, selection extension and transaction manager. Properties. |

### 4.3. UML class diagram

The following UML class diagram provides a visual representation of the types and relationships defined by this specification:

Calypso  
Networks Association

Terminal Generic Card API - 2.0.+ (2026-06-12)

- External public API references
  - Calypsonet Terminal Reader API
- Text
  - New element added since the last version
  - Element marked as deprecated since the last version
  - Element marked as deprecated
  - Work in progress...
- Object
  - Factory or interface provided by the factory
  - Interface that cannot be instantiated directly by the factory
  - SPI (Service Provider Interface) to be provided by the end-user
  - Class, enum or exception

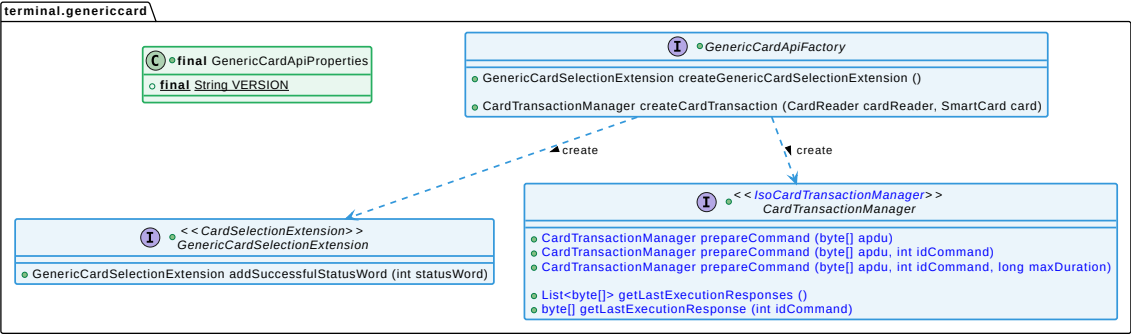


Figure 2. Terminal Generic Card API — UML Class Diagram

## 5. API Specification

### 5.1. Classes

#### 5.1.1. Generic Card API Properties

|           |                                                                    |
|-----------|--------------------------------------------------------------------|
| Name      | GenericCardApiProperties                                           |
| Kind      | Final class                                                        |
| Namespace | genericcard                                                        |
| Since     | 1.0                                                                |
| Purpose   | Exposes the immutable properties of the Terminal Generic Card API. |

#### Constants

| Name    | Type   | Since | Description                                                                                                                                                |
|---------|--------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VERSION | string | 1.0   | String representation of the version of the API implemented by the conforming binding (e.g. "2.0"). The value is a dotted decimal of the form MAJOR.MINOR. |



The `GenericCardApiProperties` type cannot be instantiated; it exposes only static properties.

### 5.2. API Interfaces

#### 5.2.1. Card Transaction Manager

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name      | CardTransactionManager                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Kind      | Interface                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Namespace | genericcard                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Extends   | IsoCardTransactionManager ( <a href="#">CNA-TR-API</a> )                                                                                                                                                                                                                                                                                                                                                                                                      |
| Since     | 1.0                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Purpose   | Provides the basic operations required to prepare, identify and process APDU exchanges with an ISO/IEC 7816-4 card. The stereotype on <code>IsoCardTransactionManager</code> grants access to the multi-channel cast <code>asMultichannelCardTransactionManager()</code> defined by <a href="#">CNA-TR-API</a> when the underlying card supports multi-channel. An instance is obtained via <u><code>GenericCardApiFactory.createCardTransaction</code></u> . |

The `CardTransactionManager` uses a **prepare/process** model inherited from [CNA-TR-API](#): the application prepares APDU commands with the `prepareCommand` overloads and processes them through the inherited `processCommands()` operation. Each prepared command MAY carry an optional **command identifier** (`idCommand`) supplied by the application, used to retrieve the corresponding response or to identify the failing command in case of error. Each prepared command MAY also carry an optional **maximum duration** (`maxDuration`, in milliseconds) used to detect a relay attack at the individual-command level.

#### Get Last Execution Response

|           |                                                                         |
|-----------|-------------------------------------------------------------------------|
| Signature | <code>getLastExecutionResponse(idCommand: integer) → byte array?</code> |
|-----------|-------------------------------------------------------------------------|

|             |                                                                                                                                                                                                                                                                                                                     |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Since       | 2.0                                                                                                                                                                                                                                                                                                                 |
| Description | Returns the response of the command identified by <b>idCommand</b> . If several prepared commands share the same identifier, the response of the <b>last</b> matching command is returned. The value returned for a given identifier is the same as the corresponding element of <b>getLastExecutionResponses</b> . |
| Parameters  | <b>idCommand</b> —the application-supplied identifier of the command whose response is requested.                                                                                                                                                                                                                   |
| Returns     | The non-empty response bytes, or <b>null</b> if no command with the supplied identifier produced a response.                                                                                                                                                                                                        |
| Throws      | None.                                                                                                                                                                                                                                                                                                               |

## Get Last Execution Responses

|             |                                                                                                                                                                                                                                                                                                                                                                 |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signature   | <code>getLastExecutionResponses() → list&lt;byte array&gt;</code>                                                                                                                                                                                                                                                                                               |
| Since       | 2.0                                                                                                                                                                                                                                                                                                                                                             |
| Description | Returns the responses collected during the last call to <b>processCommands()</b> , in the order in which the commands were prepared.<br><br>This operation does not alter the internal state of the manager: the list of responses remains available until <b>processCommands()</b> is called again, at which point it is replaced by the new set of responses. |
| Returns     | A <code>list&lt;byte array&gt;</code> representing the command responses, in the order in which the commands were sent; empty if no command has been executed.                                                                                                                                                                                                  |
| Throws      | None.                                                                                                                                                                                                                                                                                                                                                           |

## Prepare Command (APDU)

|             |                                                                                   |
|-------------|-----------------------------------------------------------------------------------|
| Signature   | <code>prepareCommand(apdu: byte array) → CardTransactionManager</code>            |
| Since       | 2.0                                                                               |
| Description | Prepares an APDU command without command identifier nor duration bound.           |
| Parameters  | <b>apdu</b> —a non-empty <code>byte array</code> containing the raw APDU command. |
| Returns     | The current instance ( <a href="#">CardTransactionManager</a> ).                  |
| Throws      | None.                                                                             |

## Prepare Command (APDU, ID Command)

|             |                                                                                                                                                                                          |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signature   | <code>prepareCommand(<br/>    apdu: byte array,<br/>    idCommand: integer<br/>) → CardTransactionManager</code>                                                                         |
| Since       | 2.0                                                                                                                                                                                      |
| Description | Prepares an APDU command with an application-supplied identifier. The identifier MAY later be used to retrieve the corresponding response via <a href="#">getLastExecutionResponse</a> . |
| Parameters  | <b>apdu</b> —a non-empty <code>byte array</code> containing the raw APDU command.<br><b>idCommand</b> —the application-supplied identifier of this command.                              |
| Returns     | The current instance ( <a href="#">CardTransactionManager</a> ).                                                                                                                         |
| Throws      | None.                                                                                                                                                                                    |

## Prepare Command (APDU, ID Command, Max Duration)

|             |                                                                                                                                                                                                                                                                                                                                                                        |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signature   | <code>prepareCommand(<br/>    apdu: byte array,<br/>    idCommand: integer,<br/>    maxDuration: long<br/>) → CardTransactionManager</code>                                                                                                                                                                                                                            |
| Since       | 2.0                                                                                                                                                                                                                                                                                                                                                                    |
| Description | Prepares an APDU command with an application-supplied identifier and a maximum tolerated exchange duration. The effective duration of the corresponding APDU exchange is measured, and an <code>InvalidCardResponseException</code> ( <a href="#">CNA-TR-API</a> ) is raised if it exceeds the declared bound; the exception message identifies the offending command. |
| Parameters  | <code>apdu</code> — a non-empty <code>byte array</code> containing the raw APDU command.<br><code>idCommand</code> — the application-supplied identifier of this command.<br><code>maxDuration</code> — the maximum tolerated duration of the APDU exchange, in milliseconds.                                                                                          |
| Returns     | The current instance ( <a href="#">CardTransactionManager</a> ).                                                                                                                                                                                                                                                                                                       |
| Throws      | None at preparation time. The duration check is performed at processing time.                                                                                                                                                                                                                                                                                          |

### 5.2.2. Generic Card API Factory

|           |                                                                                              |
|-----------|----------------------------------------------------------------------------------------------|
| Name      | <code>GenericCardApiFactory</code>                                                           |
| Kind      | Interface                                                                                    |
| Namespace | <code>genericcard</code>                                                                     |
| Since     | 1.0                                                                                          |
| Purpose   | Factory used by the application to obtain instances of the public types provided by the API. |

The `GenericCardApiFactory` is the single entry point through which the application instantiates the public types of the API. It is provided as a single instance whose accessors each return a fresh, fully initialised object.

#### Create Card Transaction

|             |                                                                                                                                                                                                                                                                        |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signature   | <code>createCardTransaction(<br/>    reader: CardReader,<br/>    card: SmartCard<br/>) → CardTransactionManager</code>                                                                                                                                                 |
| Since       | 1.0                                                                                                                                                                                                                                                                    |
| Description | Returns a new, freshly initialised <a href="#">CardTransactionManager</a> , bound to the supplied reader and to the initial card data provided by the selection process.                                                                                               |
| Parameters  | <code>reader</code> ( <code>CardReader</code> ( <a href="#">CNA-TR-API</a> )) — the reader through which the card is reached.<br><code>card</code> ( <code>SmartCard</code> ( <a href="#">CNA-TR-API</a> )) — the initial card data provided by the selection process. |
| Returns     | A <a href="#">CardTransactionManager</a> .                                                                                                                                                                                                                             |
| Throws      | None.                                                                                                                                                                                                                                                                  |

#### Create Generic Card Selection Extension

|             |                                                                                    |
|-------------|------------------------------------------------------------------------------------|
| Signature   | <code>createGenericCardSelectionExtension() → GenericCardSelectionExtension</code> |
| Since       | 1.0                                                                                |
| Description | Returns a new, freshly initialised <a href="#">GenericCardSelectionExtension</a> . |
| Returns     | A <a href="#">GenericCardSelectionExtension</a> .                                  |
| Throws      | None.                                                                              |

### 5.2.3. Generic Card Selection Extension

|           |                                                                                                                                                                                                                                                                                                                    |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name      | GenericCardSelectionExtension                                                                                                                                                                                                                                                                                      |
| Kind      | Interface                                                                                                                                                                                                                                                                                                          |
| Namespace | genericcard                                                                                                                                                                                                                                                                                                        |
| Extends   | CardSelectionExtension ( <a href="#">CNA-TR-API</a> )                                                                                                                                                                                                                                                              |
| Since     | 1.0                                                                                                                                                                                                                                                                                                                |
| Purpose   | Extends the <b>CardSelectionExtension</b> interface defined by the <b>Terminal Reader API</b> to expose a way to declare additional successful status words for the <b>Select Application</b> APDU. An instance is obtained via <u><a href="#">GenericCardApiFactory.createGenericCardSelectionExtension</a></u> . |

#### Add Successful Status Word

|             |                                                                                                                                                                                                                                                                                |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signature   | addSuccessfulStatusWord(statusWord: integer) → GenericCardSelectionExtension                                                                                                                                                                                                   |
| Since       | 1.0                                                                                                                                                                                                                                                                            |
| Description | Adds a status word to the list of those considered successful for the <b>Select Application</b> APDU.<br><b>Note:</b> the list initially contains the standard successful status word <b>9000h</b> ; each call adds to the existing set and never removes that implicit entry. |
| Parameters  | <b>statusWord</b> —a positive <b>integer</b> lower than or equal to <b>FFFFh</b> .                                                                                                                                                                                             |
| Returns     | The current instance ( <u><a href="#">GenericCardSelectionExtension</a></u> ).                                                                                                                                                                                                 |
| Throws      | None.                                                                                                                                                                                                                                                                          |